# Positive Definiteness and Carryover Sensitivity in the Hendrickson et al. (2020) Simulation Framework

*2026-10-01 18:00 PDT. Working draft.*

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**Purpose.** The simulation framework of Hendrickson et al. [1] has two known problems. First, many of the correlation matrices its data-generating process (DGP) builds are not positive definite, and the code forces them to positive definiteness by a step the paper does not describe. Second, the power of the Hybrid N-of-1 design is extremely sensitive to even very small carryover. This paper documents both problems from the published code and results, identifies a common cause, and evaluates three adjustments proposed to address them: (1) replacing compound symmetry (CS) with AR(1) within-factor correlation, (2) letting the biomarker-response correlation decay off drug, and (3) carrying the biomarker-treatment interaction in the mean of the DGP.

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## Notation index and glossary

Symbols follow `analysis/report/NOTATION.md` where it defines one (status canonical); the matrix-algebra symbols of Sections 4 to 6 and Appendix A are local to this paper. On 2026-10-09 symbols were renamed to agree with `docs/35`, `docs/37` and `analysis/report/compendium-mathematics.tex`: the coupling pattern from $u$ to $g$ ($u_i$ is the canonical random intercept), with $\tilde{g}$ its zero-padded form; the residual fraction from $g_t$ to $\phi_t$; the AR(1) step correlation from $\phi_t$ to $\rho_t$; the latent AR(1) processes of the factored generator from $P_{j,t}$ to $H_{j,t}$ ($P$ is the canonical number of randomization paths); and the normal law to $\mathcal{N}$ ($N$ is the total sample size).

(Notation, 2026-10-09: following the merged `analysis/report/NOTATION.md`, $\rho$ is the AR(1) correlation per week and the constant within-factor correlation of a compound-symmetry block is $\rho_{cs}$. Statements and formulas that belong to the compound-symmetry configurations (A and D) alone now write $\rho_{cs}$; the published parameter value 0.8 is $\rho_{cs}$ in configurations A and D and $\rho$ in the AR(1) configurations B, C, E and F, and where one value is quoted for configurations of both kinds, as in "the published parameters ($\rho = 0.8$, …)", it is both.)

Table 1: Notation index for this paper

| Symbol | Meaning | Status |
|---|---|---|
| $Y_{it}$ | observed symptom score (code `Sx`) | canonical |
| BL, $TV$, $PB$, $BR$ | baseline, time-variant, placebo-belief and biological-response components | canonical |
| $B$, $b$ | biomarker and standardized biomarker $(B - \mu_{bm})/\sigma_{bm}$ | canonical |
| $\mu_{bm}$, $\sigma_{bm}$, $\sigma_{BR}$, $\sigma_{TV}$, $\sigma_{PB}$ | biomarker mean; standard deviations of the biomarker and components | canonical |
| $c_{bm}$ | covariance-moderation parameter: on-drug biomarker-$BR$ correlation | canonical |
| $c^*_{bm}$ | ceiling: largest feasible $c_{bm}$ for a given response block and coupling pattern | canonical |
| $\beta_{bm}$ | mean-moderation multiplier (set equal to the nominal $c_{bm}$) | canonical |
| $\beta_{bm:D}$, $\widehat{\beta}_{bm:D}$ | biomarker-by-treatment interaction coefficient and its estimate (`bm:Db` in the published code) | canonical |
| $\alpha$ | nominal significance level, 0.05 | canonical |
| $N$ | total sample size across randomization paths | canonical |
| $t_{1/2}$ | carryover half-life (weeks) | canonical |
| $t_{sd}$, $t_{sd,t}$ | time since discontinuation (at visit $t$) | canonical |
| $\rho$ | AR(1) within-factor correlation per week | canonical |
| $\rho_{cs}$ | constant within-factor correlation under compound symmetry (configurations A, D) | canonical |
| $c_1$, $c_\times$ | same-visit and different-visit cross-factor correlations (code `c.cf1t`, `c.cfct`) | canonical |
| $n$, $w_t$, $d_t$ | number of visits; week of visit $t$; gap $w_t - w_{t-1}$ | canonical ($n$ local) |

| Symbol | Meaning | Status |
|---|---|---|
| $\eta_t$ | design expectancy (1 open label, 0.5 blinded), the weight scaling $\sigma_{PB}$ | canonical |
| $\phi_t$ | residual fraction of the on-drug effect at visit $t$, $2^{-t_{sd,t}/t_{1/2}}$ off drug after first exposure, 0 otherwise (graded coupling weight) | canonical |
| $g$, $g_t$, $r$ | coupling pattern (1 on drug, $\phi_t$ or $\mathbf{1}[\phi_t > 0]$ off drug) and biomarker-$BR$ correlation vector, $r = c_{bm} g$ | canonical ($r$ local) |
| $\tilde{g}$ | coupling pattern padded with zeros at the $TV$ and $PB$ entries | canonical |
| $\mu_{BR,t}$ | mean drug response at visit $t$ | canonical |
| $R$, $M$, $S$ | full correlation matrix; $3n \times 3n$ response block; diagonal matrix of standard deviations | local |
| $A$, $A_c$ | $n \times n$ within-factor (time) correlation matrix | local |
| $C_{cc'}$ | cross-factor block between factors $c$ and $c'$ | local |
| $K$, $K_a$, $K_b$, $K_o$, $K_p$ | $3 \times 3$ factor matrices in the Kronecker decompositions | local |
| $I_n$, $J_n$ | identity and all-ones $n \times n$ matrices | local |
| $\rho_t$ | one-step AR(1) correlation $\rho^{d_t}$ between consecutive visits | canonical |
| $L$, $L_A$, $L_K$ | bidiagonal factor of $A^{-1}$; Cholesky factors of $A$ and $K$ | local |
| $H_{j,t}$, $z_{j,t}$ | latent AR(1) process $j$ at visit $t$ and its standard normal innovation | local |
| $X$, $X_c$, $\mathbf{X}$, $\mathbf{Z}$ | stacked variable vector; factor $c$'s visit vector; $n \times 3$ response and standard normal matrices | local |
| $W$, $q$ | conditional-mean weights and quadratic form $[K^{-1}]_{33}\, g^\top A^{-1} g$ of the biomarker draw | local |
| $\omega_{\min}(\cdot)$, $\omega_i(\cdot)$ | smallest and $i$-th eigenvalue of a matrix | local |

**Glossary.**

- *Step coupling*: the published rule, which assigns the full $c_{bm}$ to every visit at which the carryover-adjusted $BR$ mean is nonzero.
- *Graded coupling*: the off-drug coupling $c_{bm}\phi_t$, proportional to the residual drug effect.
- *Ceiling*: the largest $c_{bm}$ for which the correlation matrix is positive definite.
- *Repair*: the published code's silent replacement of an indefinite matrix by a nearby positive definite one (`corpcor::make.positive.definite`).
- *CS*, *AR(1)*: compound symmetry (constant correlation at every lag) and first-order autoregressive correlation in calendar time.
- *Separable*: a response correlation of the form $K \otimes A$.

- *Configurations A to F*: the six data-generating configurations of Appendix A.6: A published (CS, constant cross block, step coupling); B AR(1) with the decaying cross block $c_{\times}\rho^{|w_t-w_s|}$ and step coupling; C separable AR(1) with graded coupling; D CS with the constant cross block and mean moderation; E separable AR(1) with mean moderation and no covariance coupling; F AR(1) with the decaying cross block and graded coupling. (Clarified 2026-10-09: E was labeled "Adj. 1 + 2 + 3" and "all three" in this paper. Graded coupling, adjustment 2, has no effect in E, whose coupling vector is zero; E is adjustment 1 with the separable cross block $c_1\rho^{|w_t-w_s|}$ plus adjustment 3, as arm E of `04-power-simulation.R` builds it. The labels in Appendix A.6 and Section 6 now say so.)
- *Size-adjusted power*: power computed against each cell's own null distribution rather than at nominal $\alpha = 0.05$.

## 1. Summary

1. **Problem 1 is confined to the biomarker row and is systematic.** Running the published `generateData()` at the published parameters, 40 of the 162 design-path-cell matrices fail the code's own positive definiteness check: every matrix with $c_{bm} > 0$ and no carryover in the CO, OL+BDC and Hybrid designs, plus the placebo-first CO path at every half-life. The response block itself is always valid. The failures occur because the requested biomarker-response correlation exceeds the largest value the CS structure can carry, 0.256 to 0.262 in these designs, well below the published values of 0.3 and 0.6 (Section 4).
2. **The undisclosed repair changes the simulated effect.** At $c_{bm} = 0.6$ the repair lowers the on-drug biomarker-response correlation to 0.54-0.57, introduces 0.03-0.04 at off-drug visits, and moves other correlations by up to 0.10. At $c_{bm} = 0.3$ the change is at most 0.007 (Section 4.3).
3. **Problem 2 is an artifact of the coupling rule, not of carryover.** In the published results, Hybrid power at $N = 70$, $c_{bm} = 0.3$ falls from 0.74 with no carryover to 0.12 at a half-life of 0.1 weeks, about 17 hours, when the residual drug effect one week after stopping is 0.1% of its on-drug value. The published rule sets the biomarker-response correlation to its full value wherever any residual effect remains, so with any carryover the coupling is constant across visits, the on/off contrast that identifies the interaction disappears, and the estimate collapses toward zero ($-0.28$ to $-0.07$ at $c_{bm} = 0.6$) (Section 5).
4. **The two problems share one cause.** Under the step rule, a cell without carryover has the contrast but an invalid matrix, and a cell with any carryover has a valid matrix but no contrast. The "sensitivity to small carryover" is the jump between those two regimes at a half-life of zero (Section 5.4).
5. **The three adjustments are not independent.** Decaying coupling (adjustment 2) removes the jump, but under CS it leaves the ceiling at 0.26 in every cell, so applied alone it converts problem 2 into problem 1 everywhere. AR(1) (adjustment 1) raises the ceiling where the coupling has an on/off contrast, lowers it where the coupling is constant (OL falls from 0.893 to 0.576), and does not remove the jump; on the published grid it leaves more matrices to repair, not fewer (48% against 37%). Together, with the separable AR(1) structure of `docs/35`, the Hybrid ceiling is 0.44 at the published $\rho = 0.8$: enough for $c_{bm} = 0.3$, not for 0.6. Mean moderation (adjustment 3) removes the ceiling altogether, but it is a different model of the biomarker rather than a repair of the covariance construct (Section 6).
6. **A generator that cannot simulate an invalid model.** The current package retains a second, undocumented repair in its sampler. Under the separable AR(1) structure, a factored generator draws each participant from three AR(1) recursions, a $3\times3$ mixing step and a conditional draw of the biomarker, without forming the full correlation matrix. Its conditional variance is positive exactly below the ceiling, so an infeasible effect size stops the simulation instead of being repaired. It reproduces the target matrix as closely as a standard draw does (Section 6.7).
7. **Simulation confirms both diagnoses and the remedy.** Re-simulating the published construct without any repair reproduces the collapse (Hybrid power 0.59 to 0.08 at $t_{1/2} = 0.1$ for $c_{bm} = 0.25$, `corCAR1` analysis), so the repair is not its cause. AR(1) alone leaves the collapse in place. AR(1) with graded coupling removes it: Hybrid power at $c_{bm} = 0.3$ is 0.69, 0.70 and 0.72 at half-lives of 0, 0.1 and 0.2 weeks, declining only with genuinely large carryover. Mean moderation shows no loss at any half-life (Sections 5.5 and 6.8).
8. **AR(1) data require an analysis that models serial correlation.** The published analysis assumes a constant within-participant correlation; with AR(1) data its Type I error reaches 0.14-0.23 in CO.

Adding `corCAR1` residuals restores it to 0.02-0.09. Adjustment 1 is therefore a change to the analysis as well as to the DGP (Section 6.8).

9. **Separability buys feasibility, not power.** With graded coupling, the separable cross-factor block (C) and the current `covar` block
(F) give the same power once each test is judged against its own null (within ±0.04 in Hybrid). Separability raises the ceiling (Hybrid 0.291 to 0.440), so $c_{bm} = 0.3$ becomes feasible in every cell, and it makes the response block valid for every schedule (Sections 6.4, 6.5 and 6.8).

## 2. Evidence basis

Evidence labels: **verified** (computed in this work), **inspected** (read in code or documents), **inferred** (follows from verified facts), **unverified**.

- **Published code and results.** All statements about the published framework refer to `github.com/rchendrickson/pmsi` at commit `58b32a9` (19 November 2020), the last commit before publication, here called `orig`. The repository at that commit contains the publication vignettes and the saved replicate-level results (`results_core`: 4 designs, 18 model cells, 5 censoring conditions, 1,000 replicates per cell). The data-generating code at `58b32a9` is identical to the initial commit `42ac030` (inspected, git history).
- **Problem 1** was established by running the published `generateData()` itself with the published parameters, capturing the covariance matrix passed to `mvrnorm`, and applying the published check and repair (verified). An independent reconstruction of the matrices from the specification gave identical failure patterns (verified).
- **Problem 2** was established from the published saved results (verified) and confirmed by re-simulating the published construct without repair (verified, Section 5.5).
- **Adjustments** were evaluated by exact positive-definiteness ceilings at the published parameters, and by a power simulation of six configurations, each analyzed with the published model and with serial-correlation residuals: 309,000 model fits, all converged (verified, Section 6.8; 261,000 of them from configurations A-E and 48,000 from configuration F, run later).

## 3. The published framework

Each participant's outcome is $Y_{it} = \mathrm{BL}_i - (TV_{it} + PB_{it} + BR_{it})$, with the biomarker $B_i$, baseline and the three response components drawn jointly from a multivariate normal distribution. The parameters used for the published results were (inspected, `Produce_Publication_Results_1_generate_data.Rmd`, and the saved `parameterselections`):

Table 2: Parameters of the published Hendrickson et al. simulation framework

| Parameter | Published values |
|---|---|
| Designs | OL, OL+BDC, CO, Hybrid ("N-of-1"); 8 visits over 20 weeks |
| $N$ | 35, 70 |
| $c_{bm}$ | 0, 0.3, 0.6 |
| Carryover half-life $t_{1/2}$ | 0, 0.1, 0.2 weeks |
| Within-factor correlation $\rho_{cs}$ | 0.8 for all three components, constant at every lag (CS) |
| Cross-factor correlation | $c_1 = 0.2$ same visit, $c_\times = 0.1$ otherwise |
| Biomarker-BR correlation | $c_{bm}$ wherever the carryover-adjusted BR mean is nonzero, 0 elsewhere |
| Carryover on BR mean | recursive, $\mu_{BR,t-1}(1/2)^{t_{sd}/t_{1/2}}$ with cumulative $t_{sd}$ |

| Parameter | Published values |
|---|---|
| Positive definiteness | `makePositiveDefinite = TRUE` in every call: `corpcor::make.positive.definite(sigma, tol = 1e-3)` when the check fails |
| Analysis | `lmer`, random intercept, binary on-drug indicator, test of `bm:Db` |

The Hybrid schedule places visits at weeks 4, 8, 9, 10, 11, 12, 16 and 20; all four of its paths begin on drug, as do both OL+BDC paths. CO has one path on drug first and one placebo first.

## 4. Problem 1: correlation matrices that are not positive definite

### 4.1 Incidence

Failures of the published check, by design (verified):

Table 3: Failures of the published positive definiteness check by design

| Design | Matrices checked | Failures | Where |
|---|---|---|---|
| OL | 18 | 0 | none |
| OL+BDC | 36 | 8 | both paths, $c_{bm} > 0$, $t_{1/2} = 0$ |
| CO | 36 | 16 | both paths at $t_{1/2} = 0$; the placebo-first path at every $t_{1/2}$ ($c_{bm} > 0$) |
| Hybrid | 72 | 16 | all four paths, $c_{bm} > 0$, $t_{1/2} = 0$ |
| Total | 162 | 40 | |

Every matrix with $c_{bm} = 0$ passes. Counts treat the two values of $N$ separately, as in `docs/07`, although $N$ does not enter the matrix.

### 4.2 Mechanism

The response block (all $TV$, $PB$ and $BR$ entries) is positive definite in every case: under CS with constant cross-factor terms its smallest eigenvalue is $(1 - \rho_{cs}) - (c_1 - c_\times) = 0.1$, independent of the design (verified). Failure therefore always comes from the biomarker row. With $r = c_{bm} g$ the biomarker's correlations with the $BR$ visits, the Schur complement gives the exact largest feasible value

$$c^*_{bm} = (\tilde{g}^\top M^{-1} \tilde{g})^{-1/2},$$

where $M$ is the response block and $\tilde{g}$ the coupling pattern $g$ over the $BR$ visits, padded with zeros at the $TV$ and $PB$ entries (`docs/34`, Section 3.2). For the published construct at the published parameters (verified):

Table 4: Ceiling $c^*_{bm}$ for the published construct by design and half-life

| Design | $t_{1/2} = 0$ | $t_{1/2} = 0.1$ | $t_{1/2} = 0.2$ |
|---|---|---|---|
| OL | 0.893 | 0.893 | 0.893 |
| OL+BDC | 0.262 | 0.893 | 0.893 |

| Design | $t_{1/2} = 0$ | $t_{1/2} = 0.1$ | $t_{1/2} = 0.2$ |
|---|---|---|---|
| CO | 0.256 | 0.256 | 0.256 |
| Hybrid | 0.256 | 0.893 | 0.893 |

Both published effect sizes, 0.3 and 0.6, exceed the ceiling wherever it is 0.26, which reproduces the failure pattern of Section 4.1 exactly. The ceiling is high when the coupling pattern is constant across visits (OL always; OL+BDC and Hybrid with any carryover, Section 5.3) and low when it has an on/off contrast (no carryover, or CO's placebo-first path, whose pre-exposure visits keep a zero coupling at every half-life).

### 4.3 What the repair does

When the check fails, the code replaces the covariance matrix by a nearby positive definite one, raising small and negative eigenvalues to a floor, in the spirit of nearest-positive-definite approximation [2]. The effect on the correlations actually simulated (verified):

Table 5: Correlations simulated after the published positive definiteness repair, by specified $c_{bm}$

| $c_{bm}$ specified | On-drug correlation used | Off-drug correlation used | Largest change elsewhere |
|---|---|---|---|
| 0.3 | 0.298-0.300 | 0.000-0.001 | 0.007 |
| 0.6 | 0.540-0.568 | 0.029-0.042 | 0.097 |

At $c_{bm} = 0.3$ the repair is cosmetic. At $c_{bm} = 0.6$ the simulated effect is materially smaller than specified, some interaction signal is moved into off-drug visits, and the within- and cross-factor correlations are altered as well. The published no-carryover results at $c_{bm} = 0.6$ therefore describe a DGP other than the one stated.

## 5. Problem 2: sensitivity of Hybrid power to carryover

### 5.1 The published results

Power from the published saved results, without dropout (verified):

Table 6: Published Hybrid, OL+BDC and CO power by $N$, $c_{bm}$ and half-life

| Design | $N$ | $c_{bm}$ | $t_{1/2} = 0$ | 0.1 | 0.2 |
|---|---|---|---|---|---|
| Hybrid | 35 | 0.3 | 0.54 | 0.08 | 0.07 |
| Hybrid | 35 | 0.6 | 0.95 | 0.18 | 0.23 |
| Hybrid | 70 | 0.3 | 0.74 | 0.12 | 0.10 |
| Hybrid | 70 | 0.6 | 1.00 | 0.22 | 0.33 |
| OL+BDC | 70 | 0.3 | 0.74 | 0.19 | 0.13 |
| OL+BDC | 70 | 0.6 | 0.99 | 0.36 | 0.41 |
| CO | 70 | 0.3 | 0.85 | 0.44 | 0.46 |
| CO | 70 | 0.6 | 1.00 | 0.90 | 0.91 |

Type I error at $c_{bm} = 0$ lies between 0.01 and 0.09 across cells.

### 5.2 The carryover involved is negligible

With a half-life of 0.1 weeks, the residual drug effect one week after discontinuation, the earliest off-drug visit in Hybrid and OL+BDC, is $2^{-10} \approx 0.001$ of its on-drug value; at 0.2 weeks it is $2^{-5} \approx 0.031$. A carryover this small cannot plausibly erase three quarters of the power through its effect on the response mean.

### 5.3 Mechanism: the step rule removes the contrast

The published rule assigns $c_{bm}$ to every visit at which the carryover-adjusted $BR$ mean is nonzero. Exponential and recursive carryover never reach exactly zero, so with any positive half-life every visit after first exposure receives the full $c_{bm}$. In Hybrid and OL+BDC, whose paths all begin on drug, the coupling becomes constant across all eight visits (verified, captured matrices: Hybrid path A coupling 0.6 at every visit for $t_{1/2} = 0.1$). The biomarker then predicts $BR$ equally on and off drug, which is a biomarker main effect, and the biomarker-by-treatment interaction has no population counterpart.

The published estimates show the consequence directly. The mean of the interaction estimate in Hybrid at $N = 70$ (verified):

Table 7: Mean published Hybrid interaction estimate at $N = 70$ by $c_{bm}$ and half-life

| $c_{bm}$ | $t_{1/2} = 0$ | 0.1 | 0.2 |
|---|---|---|---|
| 0.3 | −0.151 | −0.036 | −0.036 |
| 0.6 | −0.279 | −0.065 | −0.081 |

The estimate shrinks toward zero by roughly three quarters. The loss of power is primarily attenuation of the effect being estimated, not only inflation of its standard error. CO is affected less because its placebo-first path retains a zero coupling before first exposure, which preserves some contrast (and keeps that path's matrix invalid). OL has no off-drug visits, so its test does not rest on a drug contrast in the first place.

### 5.4 The common cause

Table 8: Coupling pattern, matrix validity and identification with and without carryover

| | No carryover ($t_{1/2} = 0$) | Any carryover ($t_{1/2} > 0$) |
|---|---|---|
| Coupling pattern (Hybrid, OL+BDC) | on/off contrast | constant |
| Matrix at $c_{bm} = 0.3, 0.6$ | invalid, repaired | valid |
| Interaction identified | yes | no |

The step rule makes the coupling a discontinuous function of the half-life: any positive value, however small, flips the pattern from contrast to constant. Both problems are consequences of that one feature. Problem 1 is the price of the contrast at $t_{1/2} = 0$, and problem 2 is the loss of the contrast at $t_{1/2} > 0$.

### 5.5 Confirmation by simulation

The power simulation of Section 6.8 reproduces the published construct without any repair (configuration A). Where the published code applied no repair, it matches Hendrickson's saved results: for example, Hybrid power at $c_{bm} = 0.6$ and $t_{1/2} = 0.1$ is 0.25 against the published 0.22, and OL+BDC power at the same cell is 0.36 against 0.36 (verified; agreement within about two Monte Carlo standard errors in all such cells except OL+BDC at $c_{bm} = 0.6$, $t_{1/2} = 0.2$, 0.35 against 0.41).

At $c_{bm} = 0.25$, just below the published ceiling of 0.256, the published construct is valid in every cell and is never repaired. Its Hybrid power still collapses, from 0.63 with no carryover to 0.08 at $t_{1/2} = 0.1$, and the interaction estimate falls from −0.13 to −0.03 (verified, published analysis). The collapse is therefore produced by the step rule alone, not by the repair, and it does not depend on the analysis model: with `corCAR1` residuals the same cell falls from 0.59 to 0.08.

## 6. The three adjustments

Appendix A sets out the correlation matrix each configuration builds, block by block, with a worked example at the published parameters.

### 6.1 Adjustment 1: AR(1) instead of compound symmetry

AR(1) lets correlation fall with the time between visits. At the published $\rho = 0.8$ it raises the no-carryover ceiling modestly in Hybrid and OL+BDC (0.256 to 0.291; 0.262 to 0.303) and substantially in CO (0.256 to 0.627) (verified, Section 6.5). On its own it does not change the step rule, so with any carryover the coupling is still constant and the contrast is still lost. It addresses problem 1 in part and problem 2 not at all.

(Clarified 2026-10-08: adjustment 1 as implemented here, configuration B, applies AR(1) to the cross-component block as well, $c_{\times}\rho^{|w_t - w_s|}$ (Appendix A.6), so "AR(1) alone" in this paper and in `docs/44` means AR(1) with the decaying cross block. `docs/54` uses "AR(1) only" for AR(1) with the published constant block $c_{\times}$, a different construct. With the constant block the no-carryover ceilings at $\rho = 0.8$ are 0.363 in Hybrid and 0.535 in OL, against 0.291 and 0.576 for configuration B. See `analysis/report/compendium-mathematics.tex`, Section "Ceilings by serial correlation and by cross-component form".)

### 6.2 Adjustment 2: a correlation that decays off drug

Under the graded rule the off-drug coupling is $c_{bm}\phi_t$, with $\phi_t = 2^{-t_{sd,t}/t_{1/2}}$ the residual fraction of the on-drug effect (0 before first exposure; Appendix A.4), so the coupling is a continuous function of the half-life. At $t_{1/2} = 0.1$ the residual at the first off-drug visit is 0.001, and the graded coupling is indistinguishable from the no-carryover pattern: the discontinuity of Section 5.4 is removed, and small carryover produces a small change. This addresses problem 2 directly.

Two qualifications. First, the published rule does not send the off-drug correlation to zero under carryover; it sends it to the full on-drug value. The graded rule sits between the published rule's two regimes, equal to zero-carryover behavior at negligible half-lives and decaying smoothly as the half-life grows. Second, because the graded pattern keeps the on/off contrast, it keeps the low ceiling: under CS it is 0.256-0.265 in every cell (verified). Applied alone, adjustment 2 would make $c_{bm} = 0.3$ and 0.6 infeasible in every Hybrid and OL+BDC cell, converting problem 2 into problem 1 everywhere. It must be combined with a structure that admits a larger ceiling.

### 6.3 Adjustment 3: the interaction in the mean

Under mean moderation the biomarker shifts the on-drug $BR$ mean by $\beta_{bm}\sigma_{BR}b_i$ and has no correlation with $BR$ in the matrix. The biomarker row is then trivially valid and there is no ceiling on the moderation parameter; only the response block must be positive definite, which it is under the published CS structure for every schedule (Section 4.2) and under the separable AR(1) structure of `docs/35` for every schedule (inferred from `docs/35`, Section 4.2). The interaction is carried by a mean shift that does not depend on the coupling rule, so the discontinuity of Section 5.4 does not arise. Adjustment 3 therefore addresses both problems.

It does so by changing the model rather than repairing it. Mean moderation asserts that the biomarker fixes each participant's drug effect; covariance moderation asserts that it predicts the effect only probabilistically, which is the rationale Hendrickson et al. gave for their construct [1] and the subject of paper 01 of this compendium. In paper 01's terms, adjustment 3 replaces the covariance-moderation architecture (formerly Architecture B) with the mean-moderation architecture (formerly Architecture A). It also introduces a known mismatch with an analysis model that uses a continuous exposure variable, because the mean shift is applied on drug only (paper 01, Section 2.2.1).

### 6.4 A fourth choice: separability

Adjustment 1 fixes how a factor is correlated with itself across visits (AR(1) in calendar time). It leaves open how *different* factors are correlated across visits, and that is a separate choice. The current package (`covar`) keeps Hendrickson's two cross-factor values, $c_1$ at the same visit and $c_\times$ at different visits, and lets the second decay: $c_\times \rho^{|w_t - w_s|}$. The alternative is a separable structure, in which the correlation between factor $c$ at visit $t$ and factor $c'$ at visit $s$ is a product of a factor part and a time part,

$$\mathrm{Cor}(X_{c,t}, X_{c',s}) = K_{cc'} A_{ts}, \qquad M = K \otimes A, \qquad K = (1 - c_1) I_3 + c_1 J_3,$$

so that cross-factor correlation decays at the same rate as within-factor correlation and equals $c_1$ times it at every lag. Separability is meaningful only alongside AR(1): under compound symmetry it reduces to the published construct with $c_\times = c_1 \rho_{cs}$. The full argument is in `docs/35-separable-response-covariance.md`; Appendix A.7 works it through numerically.

**What the current structure asserts.** The `covar` response block decomposes exactly as

$$M_{\text{covar}} = K_a \otimes A + K_b \otimes I_n, \qquad K_a = (1 - c_\times) I_3 + c_\times J_3, \qquad K_b = (c_1 - c_\times)(J_3 - I_3).$$

$K_b$ has a zero diagonal and positive off-diagonal entries, so it is indefinite, with eigenvalues $2(c_1 - c_\times)$ and $-(c_1 - c_\times)$. It is the same-visit excess of cross-factor correlation (0.2 rather than 0.1), and it is attached to no variance: nothing in the model generates it. Diagonalizing $A$ shows that

$$M_{\text{covar}} \succ 0 \iff \omega_{\min}(A) > \frac{c_1 - c_\times}{1 - c_\times},$$

which is 0.111 at the published values (verified, `docs/35` Section 4.4). The smallest eigenvalue of $A$ falls as visits become closer or more numerous; for equally spaced visits it stays above $(1 - \rho^d)/(1 + \rho^d)$, so at the published $\rho = 0.8$ validity at any number of visits is guaranteed only for gaps of at least one week (derived, `docs/35`). The Hybrid and OL+BDC designs, whose closest visits are one week apart, sit at that boundary.

**What separability gives.**

- **Validity for every schedule.** The eigenvalues of $K \otimes A$ are products of those of $K$ and $A$, both positive definite for any visit times, so no schedule can break the response block.
- **A larger ceiling.** At the published parameters the ceiling rises from 0.291 to 0.440 in Hybrid, from 0.303 to 0.468 in OL+BDC, and from 0.627 to 0.637 in CO (verified, Section 6.5).
- **An exact, factored ceiling and generator.** The ceiling factorizes as $c^*_{bm} = ([K^{-1}]_{33}\, g^\top A^{-1} g)^{-1/2}$, and data can be generated without forming the full matrix (Section 6.7).
- **Coherence with the analysis model.** The summed response is an AR(1) process under separability, which is the structure `corCAR1` assumes; under `covar` it is AR(1) plus a white-noise term contributed by $K_b$ (derived, `docs/35` Section 4.6).

**What it costs.** Lagged cross-factor correlations become $c_1 \rho^{\text{gap}}$ rather than $c_\times \rho^{\text{gap}}$, which doubles them at the published values, and the three factors must share one $\rho$ (the published values already do). A same-visit excess of cross-factor association cannot be represented without adding a valid occasion-level term.

Separability is therefore a fourth adjustment, distinct from the three proposed. The power simulation isolates its effect by pairing configuration C (separable) with configuration F, which differs only in using the current `covar` cross-factor block (Section 6.8).

### 6.5 The adjustments in combination

**Two ways a matrix can fail.** Either the response block $M$ (the $TV$, $PB$ and $BR$ visits) is itself not positive definite, which no choice of $c_{bm}$ can repair, or $M$ is valid and the biomarker coupling exceeds what it can support, the ceiling $c^*_{bm} = \min_{\text{paths}}(\tilde{g}^\top M^{-1}\tilde{g})^{-1/2}$ of Section 4.2. On every grid examined here the first never happens. The smallest eigenvalue of $M$, over paths and $t_{1/2} \in \{0, 0.1, 0.2, 0.5, 1\}$, is (verified):

Table 9: Smallest eigenvalue of the response block by configuration and design

| Response block | OL | OL+BDC | CO | Hybrid |
|---|---|---|---|---|
| A published (CS, constant cross) | 0.100 | 0.100 | 0.100 | 0.100 |
| B, F AR(1), `covar` cross | 0.153 | 0.010 | 0.153 | 0.010 |
| C, E AR(1), separable | 0.225 | 0.098 | 0.225 | 0.097 |

The `covar` block is valid but close to the boundary of Section 6.4 in the two designs with weekly visits. Every failure counted below is therefore a ceiling failure: the coupling, not the response structure, is what the matrix cannot hold.

**The ceilings.** Ceilings at the published parameters ($\rho = 0.8$, $c_1 = 0.2$, $c_\times = 0.1$), design-level minimum over paths (verified). Configurations as in Appendix A.6: A published (CS, constant cross-factor, step coupling); B adjustment 1 (AR(1), step); F adjustments 1 + 2 with the `covar` cross-factor block; C adjustments 1 + 2, separable. D and E (mean moderation) have no ceiling.

Table 10: Ceilings for configurations A, B, F and C by design and half-life

| Config. | Design | $t_{1/2} = 0$ | 0.1 | 0.2 | 0.5 | 1 |
|---|---|---|---|---|---|---|
| A | OL | 0.893 | 0.893 | 0.893 | 0.893 | 0.893 |
| A | OL+BDC | 0.262 | 0.893 | 0.893 | 0.893 | 0.893 |
| A | CO | 0.256 | 0.256 | 0.256 | 0.256 | 0.256 |
| A | Hybrid | 0.256 | 0.893 | 0.893 | 0.893 | 0.893 |
| B | OL | 0.576 | 0.576 | 0.576 | 0.576 | 0.576 |
| B | OL+BDC | 0.303 | 0.594 | 0.594 | 0.594 | 0.594 |
| B | CO | 0.627 | 0.576 | 0.576 | 0.576 | 0.576 |
| B | Hybrid | 0.291 | 0.595 | 0.595 | 0.595 | 0.595 |
| F | OL | 0.576 | 0.576 | 0.576 | 0.576 | 0.576 |
| F | OL+BDC | 0.303 | 0.303 | 0.314 | 0.404 | 0.517 |
| F | CO | 0.627 | 0.627 | 0.627 | 0.627 | 0.627 |
| F | Hybrid | 0.291 | 0.291 | 0.302 | 0.386 | 0.486 |
| C | OL | 0.567 | 0.567 | 0.567 | 0.567 | 0.567 |
| C | OL+BDC | 0.468 | 0.469 | 0.476 | 0.526 | 0.578 |
| C | CO | 0.637 | 0.637 | 0.637 | 0.637 | 0.637 |
| C | Hybrid | 0.440 | 0.440 | 0.446 | 0.491 | 0.544 |

Adjustment 2 alone (CS, graded coupling) was computed separately for $t_{1/2} \leq 0.2$: 0.256 / 0.256 / 0.257 in Hybrid, 0.262 / 0.262 / 0.265 in OL+BDC and 0.256 in CO (verified, `03-adjustment-ceilings.R`). The ceilings predict the repair counts below exactly: a path's matrix fails the published test if and only if $c_{bm}$ exceeds that path's ceiling, for all 504 matrices counted (verified).

**Why a ceiling must exist.** The quantity $c^2_{bm}\,\tilde{g}^\top M^{-1}\tilde{g}$ is the squared multiple correlation of the biomarker on the whole response trajectory, so it cannot reach 1. A single biomarker cannot be correlated 0.6 with each of several on-drug visits that are only partly correlated with one another: jointly those correlations would account for more than all of its variance. The bound tightens with every additional on-drug visit that carries partly independent noise. This is why AR(1) lowers the OL ceiling from 0.893 to 0.567-0.576. Under compound

symmetry the visits share a large person-level component that the biomarker can correlate with once; under AR(1) far less is shared, so the same correlation at every visit demands more of the biomarker. The ceiling is therefore a property of encoding the interaction as a correlation, not a defect of any one construct; better structure raises it, and only moving the interaction into the mean removes it.

**The step rule's high ceilings are the cells without signal.** Under the step rule, the Hybrid and OL+BDC ceilings jump from about 0.26 to 0.893 (A) or 0.595 (B) as soon as $t_{1/2} > 0$. The coupling becomes constant after first exposure, and a constant vector pays almost no switch cost (Appendix A.7.4). It also carries no on-drug against off-drug contrast, which is the collapse of Section 5.3. In the published construct, the cells that stop needing repair are the cells that lose their signal.

**Repair counts.** The share of correlation matrices that the published code would silently repair, counted with its own test (`corpcor::is.positive.definite()`), one matrix per design path × $c_{bm}$ × $t_{1/2}$ (verified). For the strawman the published grid uses the matrices of the 58b32a9 code itself; configuration A reproduces them exactly (0 of 81 disagree), and a fresh run of the 58b32a9 code on the Figure 4 grid (`10-strawman-58b32a9.R`, 100 replicates) logged the same 20 repairs during the run.

Table 11: Share of matrices repaired by configuration on the published and common grids

| Configuration | Published grid, $c_{bm}$ 0.3 and 0.6 | Common grid, $c_{bm}$ 0.25 and 0.3 |
|---|---|---|
| Strawman (58b32a9 code) | 20/54 (37%) | 12/90 (13%) |
| B AR(1) | 26/54 (48%) | 3/90 (3%) |
| F AR(1) + graded, `covar` | 26/54 (48%) | 5/90 (6%) |
| C AR(1) + graded, separable | 21/54 (39%) | 0/90 (0%) |
| D mean moderation | 0/54 (0%) | 0/90 (0%) |
| E separable AR(1) + mean moderation | 0/54 (0%) | 0/90 (0%) |

The published grid is that of Figure 4 (OL, OL+BDC, CO, Hybrid; $t_{1/2} \in \{0, 0.1, 0.2\}$). The common grid is that of the strawman-against-fixes contrast run through the 58b32a9 pipeline (`12-common-grid-contrast.R`; same designs, $N = 70$, $t_{1/2} \in \{0, 0.1, 0.2, 0.5, 1\}$; its power results are not yet reported here). In the common grid every strawman repair is at $c_{bm} = 0.3$: all CO cells, and the $t_{1/2} = 0$ cells of OL+BDC and Hybrid. None is at 0.25.

Read against the published effect sizes:

- $c_{bm} = 0.3$ is feasible in every cell only with adjustments 1 and 2 together in the separable form, or with adjustment 3. The current `covar` form of 1 + 2 is marginal in Hybrid (0.291 against 0.3).
- $c_{bm} = 0.6$ exceeds every covariance-moderation ceiling in Hybrid and OL+BDC at $\rho = 0.8$. No covariance construct examined can represent the published large effect in those designs; only mean moderation can. At the project's $\rho = 0.7$ (no carryover) the separable ceiling is 0.480 in Hybrid (path A; 0.491 on path C) and 0.474 in OL+BDC (`docs/35`, Section 6), still below 0.6.

### 6.6 Which adjustment addresses which problem

Table 12: Which adjustment addresses each of the two problems

| | Problem 1 (invalid matrices) | Problem 2 (carryover sensitivity) |
|---|---|---|
| 1. AR(1) | No, on balance: raises the ceiling where the coupling has contrast (CO; Hybrid and OL+BDC without carryover), lowers it where the coupling is constant (OL; carryover cells); 48% of published-grid matrices fail against 37% | No: the step remains |
| 2. Graded coupling | No: alone it lowers the ceiling in carryover cells to the no-carryover level | Yes: removes the discontinuity |
| 1 + 2 (separable) | Yes for $c_{bm} \leq 0.44$ (Hybrid, $\rho = 0.8$) | Yes |
| 3. Mean moderation | Yes: no ceiling | Yes, by a different model |

The problem-2 entries were first established from the structure of the coupling and the ceiling, and are confirmed by the power simulation of Section 6.8.

### 6.7 Implementation: a factored generator

Both problems were possible because the published code could simulate a model that does not exist: when the requested correlation matrix was not positive definite, it was repaired and sampled without notice. The current package keeps that possibility in a second form. Its sampler factors the covariance matrix once and caches the factor, and if the factorization fails it repairs the matrix and continues, whether or not `makePositiveDefinite` is set (inspected: `R/generateData.R:391-393`, with identical fallbacks in `implementations/tidyverse`, `implementations/nof1power` and `implementations/original-extended`). Removing the documented repair flag is therefore not enough.

Under the separable AR(1) structure of adjustments 1 and 2, a participant can instead be generated in three stages that never form or factor the full $(2+3n) \times (2+3n)$ correlation matrix, and in which an infeasible effect size cannot be simulated.

**1. Time.** Draw three independent AR(1) paths across the visits by recursion on the gaps $d_t = w_t - w_{t-1}$:

$$H_{j,1} \sim \mathcal{N}(0,1), \qquad H_{j,t} = \rho_t H_{j,t-1} + \sqrt{1-\rho_t^2}\, z_{j,t}, \qquad \rho_t = \rho^{d_t}, \qquad j = 1,2,3,$$

with $z_{j,t}$ independent standard normal innovations.

Each path has correlation $\rho^{|w_t - w_s|}$ between visits, for any spacing [3, 4].

**2. Factors.** Mix the paths with the Cholesky factor of the $3 \times 3$ factor correlation $K = (1-c_1)I_3 + c_1 J_3$:

$$(TV_t, PB_t, BR_t)^\top = L_K\,(H_{1,t}, H_{2,t}, H_{3,t})^\top, \qquad L_K L_K^\top = K.$$

The response block then has correlation $K \otimes A$ exactly, because $\mathrm{chol}(K \otimes A) = \mathrm{chol}(K) \otimes \mathrm{chol}(A)$ [5].

**3. Biomarker.** Draw the biomarker given the response:

$$B \mid X \sim \mathcal{N}\Big(c_{bm} \sum_c [K^{-1}]_{c3}\, X_c^\top A^{-1} g,\;\; 1 - c_{bm}^2\, q\Big), \qquad q = [K^{-1}]_{33}\, g^\top A^{-1} g,$$

where $X_c$ is factor $c$'s vector of visits and $g$ the coupling pattern of Appendix A.4. $A^{-1}g$ is computed from the bidiagonal AR(1) precision, $A^{-1} = L^\top L$ with $L_{11} = 1$, $L_{tt} = (1-\rho_t^2)^{-1/2}$ and $L_{t,t-1} = -\rho_t(1-\rho_t^2)^{-1/2}$, in a few multiplications; the only matrix inverted is the $3 \times 3$ $K$. The baseline is drawn independently. Means, standard deviations, carryover on the mean and, under mean moderation, the on-drug shift are applied after the draw, as in the existing generator.

**The ceiling is the draw.** The biomarker's conditional variance $1 - c_{bm}^2 q$ is positive exactly when $c_{bm}$ is below the ceiling $c_{bm}^* = q^{-1/2}$ of Section 4.2. If it is not, there is no valid distribution to draw from, and the generator stops and reports the ceiling rather than repairing anything.

**Checks** (verified, `analysis/scripts/quick-sim/cbm-ceiling/07-factored-generator.R`; Hybrid path A, graded coupling at $t_{1/2} = 0.5$, $\rho = 0.8$, $c_1 = 0.2$, $c_{bm} = 0.3$):

Table 13: Checks of the factored generator against the full-matrix computation

| Check | Result |
|---|---|
| Ceiling, closed form against full matrix | 0.490934 against 0.490934 |
| $A^{-1}g$, closed form against `solve()` | agree to $8 \times 10^{-16}$ |
| Sampled correlations against the target, 5 seeds of 400,000 draws | largest deviation 0.0034-0.0056 |
| Same check for a standard full-matrix draw | largest deviation 0.0037-0.0052 |
| $c_{bm} = 0.6$ requested | refused: "exceeds the ceiling 0.491" |

(Corrected 2026-10-08: the check script entered the coupling at visit 8 (week 20, $t_{sd} = 4$) as 0.0625; graded coupling at $t_{1/2} = 0.5$ gives $2^{-8} = 0.0039$ there. With the source's vector the ceiling was 0.494329; with the correct vector it is 0.490934, which agrees with the configuration C entry for Hybrid at $t_{1/2} = 0.5$ (0.491) in the Section 6.5 table. The script was corrected and rerun; the factored-draw deviations were unchanged, the full-matrix deviations changed from 0.0035-0.0060, the $A^{-1}g$ agreement from $7 \times 10^{-16}$, and the refusal message from 0.494, and the conclusion is unchanged. See `analysis/report/compendium-mathematics.tex`, Section "Silent repair, and why the generator must stop".)

The factored draws reproduce the target correlation matrix as closely as a standard full-matrix draw does.

**Advantages.**

- **An invalid model cannot be simulated.** The validity check is part of generation, so the silent repair of both problems has no route back in (verified).
- **The ceiling is known before simulating** and can be reported with every power estimate (verified, closed form).
- **Each stage is a stated assumption:** one AR(1) clock over time, a fixed blend across factors, the biomarker given the response. The model is readable from the code and testable piece by piece (inferred).
- **Common random numbers across effect sizes.** The response block does not depend on $c_{bm}$, so the same response trajectories can be reused for every $c_{bm}$ in a grid, with only the biomarker redrawn; comparisons across effect sizes then differ by the effect alone (inferred; not implemented).
- **Any schedule, and long designs.** The recursion is valid for any gaps, and its cost grows linearly with the number of visits rather than with the cube of the matrix size. For eight visits this makes no practical difference; for designs with many visits it would (inferred, not timed).

**Limits.** The factored generator exists only for the separable AR(1) structure with a common $\rho$; the current `covar` (separate $c_\times$) and the published compound-symmetry construct cannot be generated this way. With the same seed it produces different individual participants from the existing generator, so existing results are reproducible in distribution but not draw for draw. It has been checked against its target distribution, but not yet integrated into `buildSigma()` or used in a power simulation.

### 6.8 Power under the adjustments

**Design.** The six configurations of Appendix A.6 were simulated with everything else held at the published values: $\rho = 0.8$, $c_1 = 0.2$, $c_\times = 0.1$, the published Gompertz means and standard deviations, the published recursive carryover on the $BR$ mean, and the three designs with $N = 70$ allocated across paths as in the published code. The grid crossed $c_{bm} \in \{0, 0.25, 0.3, 0.6\}$ with $t_{1/2} \in \{0, 0.1, 0.2, 0.5, 1\}$ weeks, 500 replicates per cell, giving a Monte Carlo standard error of about 0.022 on power near 0.5 and 0.010 on a Type I error of 0.05. The value $c_{bm} = 0.25$ lies just below the published ceiling, so it is feasible in every configuration. Cells whose matrix is not positive definite were skipped rather than repaired (51 of 360, exactly those predicted by the ceilings of Section 6.5). Each replicate was analyzed twice, with identical data: by the published model (random intercept, binary on-drug indicator, test of `bm:Db`), and by the same model with continuous-time AR(1) residuals (`nlme::lme` with `corCAR1`). All 309,000 fits converged (verified). Configuration F was run after A-E, with its own seeds; the seeds of A-E were not changed.

**Calibration of the analysis.** The published analysis assumes the within-participant correlation is constant, which is exactly right for compound-symmetry data and wrong for AR(1) data. With AR(1) data it rejects a true null too often: Type I error reaches 0.14-0.23 in CO and 0.04-0.10 in Hybrid and OL+BDC for configurations B, C, E and F, against 0.03-0.06 in CO for the compound-symmetry configurations A and D (Figure 1, verified). With `corCAR1` residuals, Type I error in the AR(1) configurations falls to 0.02-0.09, close to nominal; F is the most conservative of them in Hybrid and OL+BDC (0.016-0.052). Adopting AR(1) data (adjustment 1) therefore requires an analysis that models serial correlation; the results below use the `corCAR1` analysis. One pattern is not explained by this: in OL+BDC, the compound-symmetry configurations A and D run at 0.06-0.11 under both analyses.

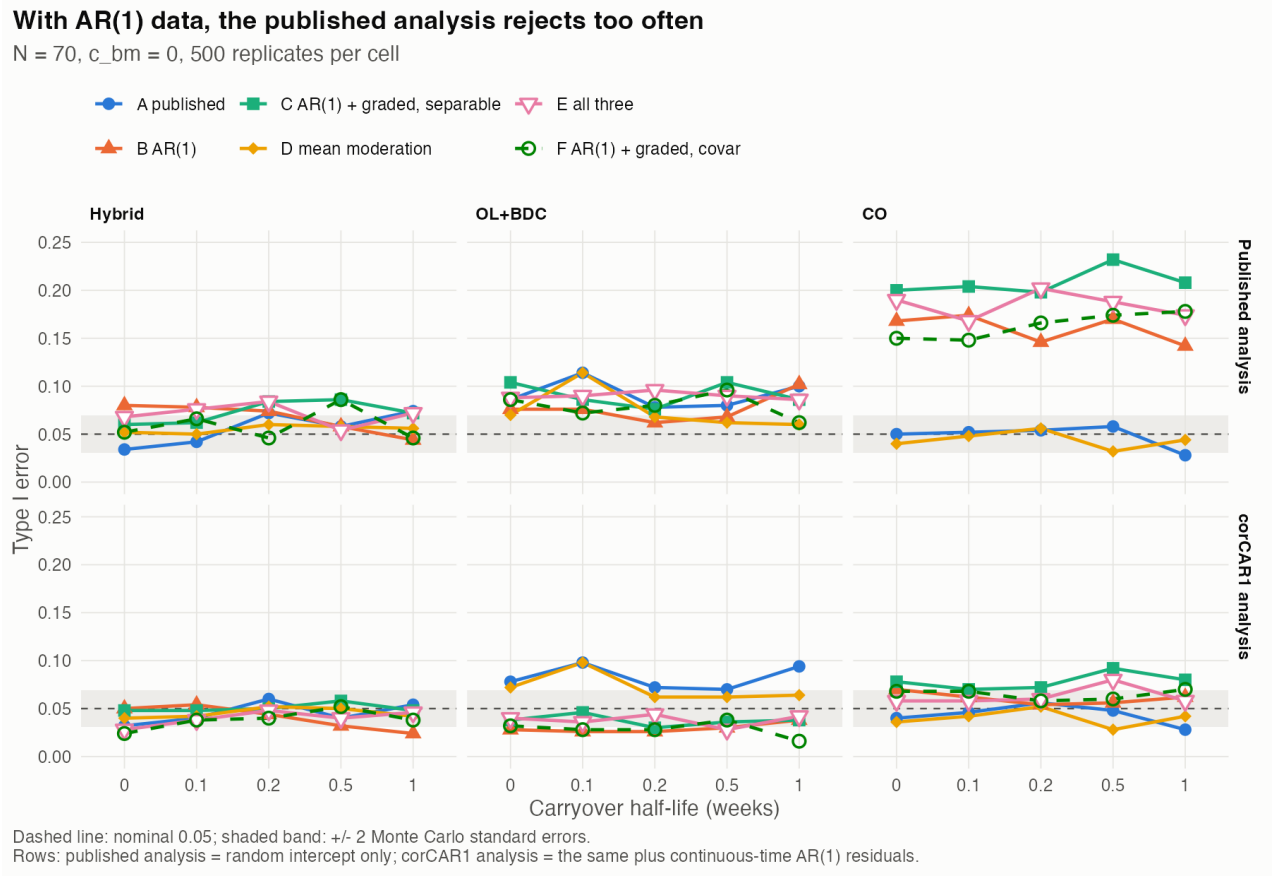



Figure 1: Type I error ($c_{bm} = 0$) by configuration, design and carryover half-life, under the published analysis (top) and the `corCAR1` analysis (bottom). Shaded band: $\pm 2$ Monte Carlo standard errors around 0.05.

**Power.** Nominal power in the Hybrid design, `corCAR1` analysis (verified):

Table 14: Nominal Hybrid power by configuration, $c_{bm}$ and half-life, corCAR1 analysis

| Configuration | $c_{bm}$ | $t_{1/2} = 0$ | 0.1 | 0.2 | 0.5 | 1 |
|---|---|---|---|---|---|---|
| A Published | 0.25 | 0.59 | 0.08 | 0.08 | 0.09 | 0.10 |
| B AR(1) | 0.25 | 0.44 | 0.06 | 0.06 | 0.06 | 0.06 |
| F AR(1) + graded, `covar` | 0.25 | 0.43 | 0.41 | 0.44 | 0.36 | 0.24 |
| F AR(1) + graded, `covar` | 0.30 | infeasible | infeasible | 0.59 | 0.46 | 0.34 |
| C AR(1) + graded, separable | 0.25 | 0.51 | 0.48 | 0.52 | 0.41 | 0.28 |
| C AR(1) + graded, separable | 0.30 | 0.69 | 0.70 | 0.72 | 0.55 | 0.36 |
| D Mean moderation | 0.30 | 0.78 | 0.75 | 0.74 | 0.77 | 0.73 |
| E Separable AR(1) + mean moderation | 0.30 | 0.67 | 0.65 | 0.67 | 0.66 | 0.64 |

Five findings follow.

1. **AR(1) alone does not help.** Configuration B collapses as A does, to 0.06 at $t_{1/2} = 0.1$ in Hybrid, because it keeps the step rule.
2. **Graded coupling removes the collapse.** Under configuration C, power at $t_{1/2} = 0.1$ and 0.2 equals power without carryover (0.69, 0.70, 0.72 at $c_{bm} = 0.3$), as Section 6.2 predicted. It then declines with genuinely large carryover (0.55 at $t_{1/2} = 0.5$, 0.36 at 1 week). That decline is real rather than artifactual: at those half-lives the off-drug visits carry part of the interaction, which a binary on-drug indicator codes as absent, and the estimate attenuates accordingly (from −0.16 to −0.11; Figure 3).
3. **Mean moderation is flat.** Configurations D and E show no loss at any half-life, and their estimates sit at the expected value throughout. Under mean moderation the interaction exists only on drug, which is exactly what the binary indicator codes, so the analysis is correctly aligned with the DGP. (Paper 01's mismatch for mean moderation arises from its continuous exposure variable, not from mean moderation itself.)
4. **The response structure matters for CO.** CO power at $c_{bm} = 0.3$ is 0.85 under D but 0.44 under E, which differ only in compound-symmetry against separable AR(1) responses. Under compound symmetry the on-drug and off-drug blocks remain correlated at 0.8 and the participant-level noise cancels from the contrast; under AR(1), with 2.5-week spacing, they are nearly independent. Hybrid is far less affected (0.78 against 0.67).
5. **Separability changes feasibility, not power.** Configurations F and C differ only in the cross-factor block (Section 6.4). Both remove the collapse, and their curves have the same shape (Figure 2). Nominal Hybrid power is higher under C by 0.04-0.08 at $c_{bm} = 0.25$, about one to two and a half standard errors of a difference (0.031). Most of that gap is a calibration effect: F's `corCAR1` test is the more conservative (Type I 0.024-0.052 against 0.048-0.058 for C in Hybrid). Against each cell's own null the two are within ±0.04 in every Hybrid cell (C 0.52, 0.50, 0.53, 0.37, 0.29; F 0.56, 0.47, 0.51, 0.35, 0.27), and OL+BDC and CO show no consistent ordering either. What separability does change is the feasible range: at $c_{bm} = 0.3$, F cannot be run in Hybrid for $t_{1/2} \leq 0.1$ (ceiling 0.291) while C can (0.440), and C's response block is valid for every schedule. Its case therefore rests on structure (Section 6.4), not on power.

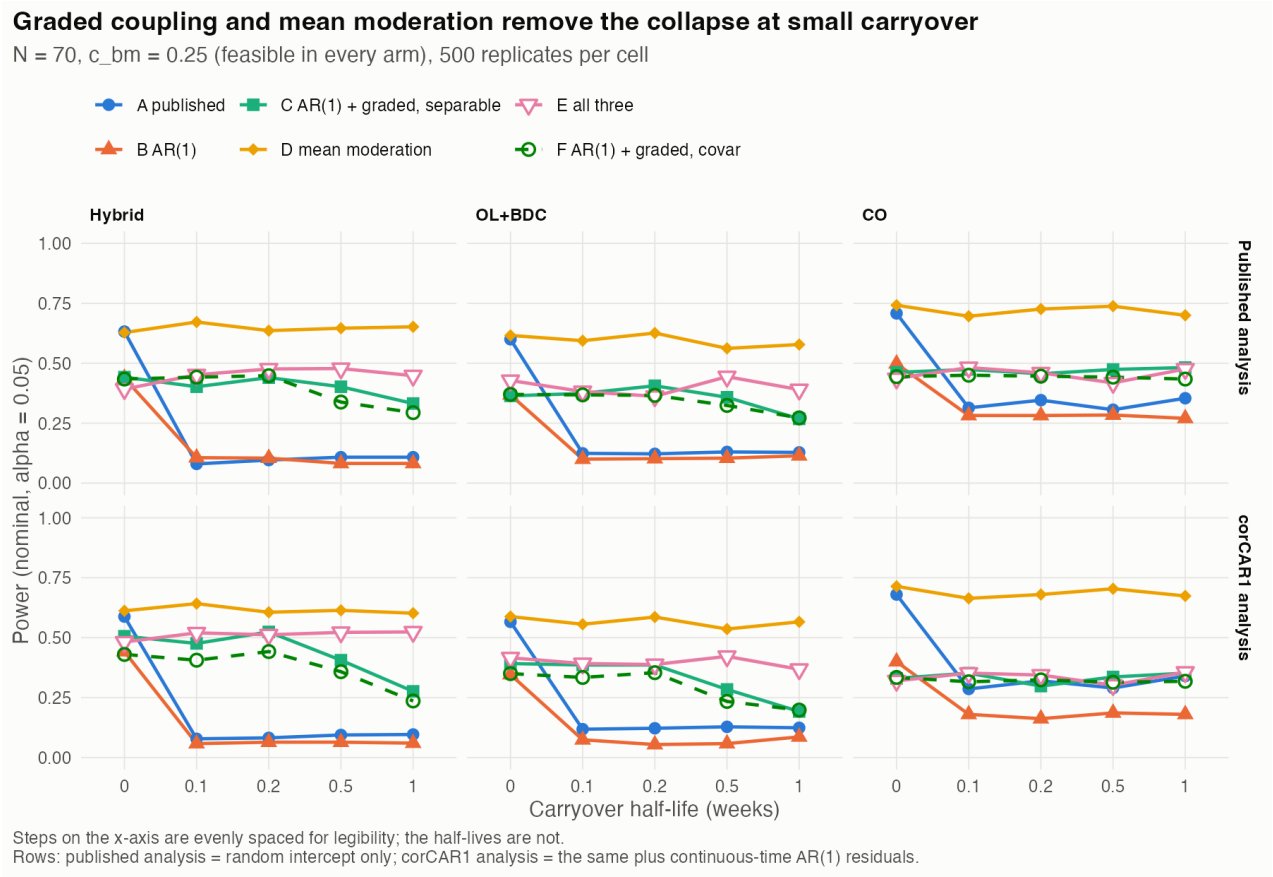



Figure 2: Power at $c_{bm} = 0.25$ by configuration, design and carryover half-life, under the published (top) and `corCAR1` (bottom) analyses. Steps on the horizontal axis are evenly spaced; the half-lives are not.

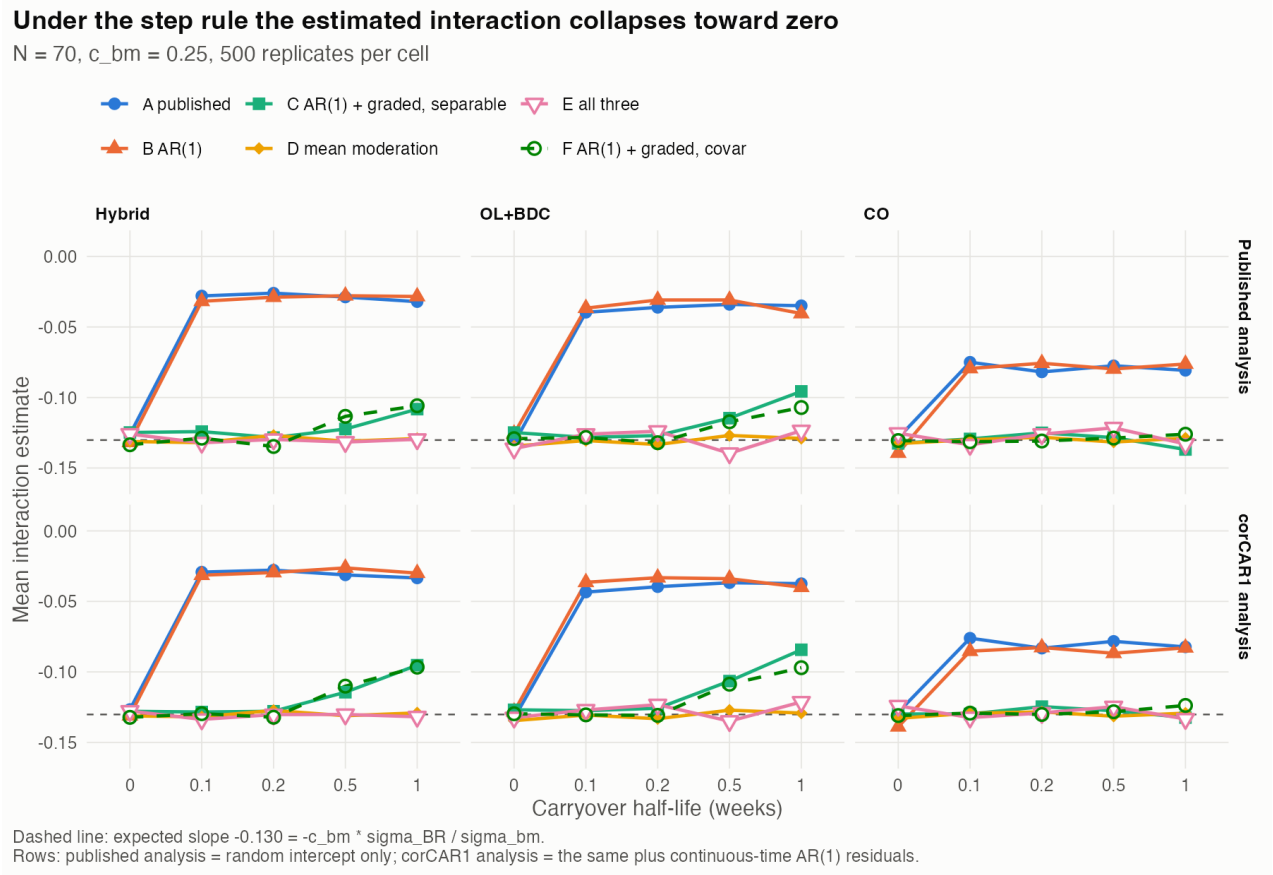



Figure 3: Mean interaction estimate $\hat{\beta}_{bm:D}$ at $c_{bm} = 0.25$, with the expected slope $-c_{bm}\sigma_{BR}/\sigma_{bm} = -0.130$ (dashed). Under the step rule (A, B) the estimate collapses toward zero with any carryover; under graded coupling (C) it holds until the carryover is large; under mean moderation (D, E) it does not move.

**Size adjustment.** Power computed against each cell's own null distribution, rather than at the nominal 0.05, differs from nominal power where Type I error departs from 0.05, by up to 0.15 (configuration A, Hybrid, no carryover: 0.59 nominal against 0.74 adjusted, with Type I error 0.032). The five findings above hold under size adjustment: in Hybrid at $c_{bm} = 0.3$, configuration C runs 0.70, 0.71, 0.72, 0.52 and 0.37 across the half-lives, D runs 0.74-0.81 throughout, and E 0.65-0.75; at $c_{bm} = 0.25$, configuration A falls from 0.74 to 0.09 (verified; `power-summary-reps500-car1.csv`).

## 7. Implications

- **For the published results.** The no-carryover cells with $c_{bm} > 0$ in OL+BDC, CO and Hybrid, and the placebo-first CO path throughout, were generated from repaired matrices; at $c_{bm} = 0.6$ the simulated effect was smaller than stated (inferred from Section 4.3). The large loss of Hybrid and OL+BDC power with carryover is a property of the coupling rule, not of carryover, and should not be read as evidence that these designs are fragile to realistic carryover (inferred from Section 5).
- **For the project's DGP.** Adopt adjustments 1 and 2 together, in the separable form, as the covariance-moderation DGP, generated with the factored generator of Section 6.7. That removes both silent repairs, the documented flag and the undocumented fallback in the sampler, so that an infeasible effect size is an error rather than a quietly different simulation. Choose effect sizes below the ceiling, and report the ceiling alongside power.
- **For the analysis.** AR(1) data analyzed without a serial-correlation term give anti-conservative tests (Section 6.8). Any simulation that adopts adjustment 1 should analyze with `corCAR1` (or an equivalent) residuals, and report Type I error alongside power.
- **On large effects.** If effect sizes near 0.6 must be studied in Hybrid or OL+BDC, the covariance construct cannot carry them, and the choice is between mean moderation (adjustment 3), with its different biological commitment, and a smaller within-factor correlation.

## 8. Relation to earlier project documents

- `docs/07` (positive definiteness failures) reports the same total of 40 failures in 162 checks, but its breakdown by design (32 of 40 in OL+BDC, 2 of 24 in Hybrid) and its explanation through the expectancy scaling of the placebo variance do not reproduce: positive diagonal scaling cannot change positive definiteness, and the published code's own matrices fail as in Section 4.1. Superseded by Section 4.
- `docs/07a` gives an empirical ceiling of about 0.6 under an AR(1) structure the published code does not use. Superseded.
- `docs/09` (carryover correlation artifact) correctly identifies the step rule, but attributes the loss to standard-error inflation with an unbiased estimate; the published estimates attenuate by about three quarters (Section 5.3). It also contains arithmetic slips in the residual fractions.
- `docs/13` (Figure 4 walkthrough) labels its code as the initial commit but describes the 2024 revision's coupling, scale factor and analysis variable.
- `docs/32` and `docs/34`-`docs/35` are consistent with this paper and supply the ceiling formula and the separable structure.
- *Note added 2026-10-07.* The later `docs/37` (closed-form carryover power) derives the same result analytically: carryover acts on interaction power only through the biomarker coupling gap between on-drug and off-drug visits, the step rule makes that gap zero at any positive half-life, and graded coupling ($\phi_t$ off drug, 0 before first exposure) removes the collapse.

## 9. Open work

The power simulation of Section 6.8 completed the original item 1 (power under the adjustments, with the published and a serial-correlation analysis) and settled the question of repair against flattening: the collapse occurs without repair (Section 5.5). Remaining:

1. **Continuous exposure in the analysis.** Section 6.8 uses the binary on-drug indicator throughout. Configuration C's decline at large half-lives reflects that indicator coding partial off-drug interaction as absent; a continuous exposure variable (`Dbc`) should recover it, and under mean moderation should introduce paper 01's mismatch instead. Both are predictions, not yet tested.
2. **The project's other changes.** The project DGP also corrects the recursive carryover on the mean and uses $\rho = 0.7$. Each should be added one at a time to configuration C.
3. **Smaller samples and other designs.** Only $N = 70$ and the three published designs were simulated; the published $N = 35$ and multi-cycle designs, where the ceiling is lowest, remain.
4. **Unexplained Type I error in OL+BDC.** The compound-symmetry configurations run at 0.06-0.11 in OL+BDC under both analyses (Section 6.8), which neither analysis model accounts for.
5. **The factored generator in the package.** Integrate Section 6.7 into `buildSigma()`, and confirm by simulation that it reproduces configuration C's power.

## 10. Limitations

- The ceilings and failure counts are exact for the published designs and parameters; other designs and parameter values will differ.
- The repair was applied to the covariance matrix as in the published code, and its effect is reported on the correlation scale.
- The power simulation used 500 replicates per cell (Monte Carlo standard error about 0.022 on power near 0.5), one sample size ($N = 70$), the published parameters, and a binary on-drug indicator in the analysis. Differences between configurations smaller than about 0.06 should not be read as real.
- It ran in the host R session rather than the project container.
- Reference [2] describes the class of method; the exact algorithm of `corpcor::make.positive.definite` was characterized here by its effect, not by source review.

## 11. Reproducibility

```
bash analysis/scripts/quick-sim/hendrickson-problems/00-fetch-58b32a9.sh
Rscript analysis/scripts/quick-sim/hendrickson-problems/01-published-matrices.R
Rscript analysis/scripts/quick-sim/hendrickson-problems/02-published-power.R
Rscript analysis/scripts/quick-sim/hendrickson-problems/03-adjustment-ceilings.R
Rscript analysis/scripts/quick-sim/hendrickson-problems/06-worked-example-matrices.R
Rscript analysis/scripts/quick-sim/hendrickson-problems/08-separable-factorization-example.R
Rscript analysis/scripts/quick-sim/hendrickson-problems/11-pd-repair-count.R
Rscript analysis/scripts/quick-sim/cbm-ceiling/07-factored-generator.R
Rscript analysis/scripts/quick-sim/hendrickson-problems/04-power-simulation.R --reps 500 --analysis publ
Rscript analysis/scripts/quick-sim/hendrickson-problems/04-power-simulation.R --reps 500 --analysis car1
Rscript analysis/scripts/quick-sim/hendrickson-problems/04-power-simulation.R --reps 500 --analysis publ
Rscript analysis/scripts/quick-sim/hendrickson-problems/04-power-simulation.R --reps 500 --analysis car1
Rscript analysis/scripts/quick-sim/hendrickson-problems/05-summarize-power.R --reps 500 --analysis publi
Rscript analysis/scripts/quick-sim/hendrickson-problems/05-summarize-power.R --reps 500 --analysis car1
Rscript analysis/scripts/quick-sim/hendrickson-problems/07-power-figures.R
```

Run from the repository root. The first script fetches the `58b32a9` sources and saved results with the GitHub CLI into `analysis/data/quick-sim/hendrickson-58b32a9/`; the others write their tables to `analysis/data/quick-sim/hendrickson-problems/`. The power simulation takes about 25 minutes with the published analysis and about 60 minutes with `corCAR1` on 8 cores; it saves each cell as it finishes and resumes from those files if interrupted. Fixed per-cell seeds give both analyses identical data. Configuration F was run afterwards with `--arms F`; its cells keep their place in the full grid, so its seeds do not disturb those of A-E, and the summary script merges the two sets of output.

## 12. References


1. Hendrickson RC, Thomas RG, Schork NJ, Raskind MA. Optimizing aggregated N-of-1 trial designs for predictive biomarker validation: statistical methods and theoretical findings. *Frontiers in Digital Health* 2020;2:13. doi:10.3389/fdgth.2020.00013. (Verified in PubMed, PMID 34713026.) Code: `github.com/rchendrickson/pmsimstats`, commit `58b32a9`.
2. Higham NJ. Computing a nearest symmetric positive semidefinite matrix. *Linear Algebra and its Applications* 1988;103:103-118. doi:10.1016/0024-3795(88)90223-6. (Verified in Crossref.)
3. Diggle PJ. An approach to the analysis of repeated measurements. *Biometrics* 1988;44(4):959-971. (Verified in PubMed, PMID 3233259.)
4. Jones RH, Boadi-Boateng F. Unequally spaced longitudinal data with AR(1) serial correlation. *Biometrics* 1991;47(1):161-175. (Verified in PubMed, PMID 2049497.)
5. Horn RA, Johnson CR. *Topics in Matrix Analysis*. Cambridge University Press; 1991. doi:10.1017/CBO9780511840371. (Verified in Crossref.)


## Appendix A. Matrix representations of the data-generating processes

This appendix sets out the correlation matrix that each configuration in this paper assembles, so that the differences discussed in Sections 4 to 6 can be read off directly. It adapts Appendix B of paper 01 (`analysis/report/01-dgp-mean-moderation-vs-mvn/report.Rmd`) in two ways: it covers all six configurations of the power simulation (Section 6.8), including the separable form and mean moderation, and its worked example uses the published parameters ($\rho = 0.8$, $c_1 = 0.2$, $c_\times = 0.1$) rather than paper 01's illustrative values. The participant subscript is suppressed throughout.

### A.1 Variable ordering and block partition

Each participant's trajectory is one draw from a multivariate normal distribution over the ordered variables

$$X = (B,\ \mathrm{BL},\ TV_1, \dots, TV_n,\ PB_1, \dots, PB_n,\ BR_1, \dots, BR_n)^\top, \qquad \dim X = 2 + 3n,$$

for $n$ visits at cumulative weeks $w_1 < \cdots < w_n$. The ordering is factor-major: all visits of one factor are contiguous. Writing $A_c$ for the $n \times n$ within-factor block of factor $c \in \{tv, pb, br\}$, $C_{cc'}$ for the cross-factor block, and $r$ for the vector of biomarker-$BR$ correlations, the correlation matrix is

$$R = \begin{pmatrix} 1 & 0 & 0^\top & 0^\top & r^\top \\ 0 & 1 & 0^\top & 0^\top & 0^\top \\ 0 & 0 & A_{tv} & C_{tv,pb} & C_{tv,br} \\ 0 & 0 & C_{pb,tv} & A_{pb} & C_{pb,br} \\ r & 0 & C_{br,tv} & C_{br,pb} & A_{br} \end{pmatrix}.$$

Two features hold in every configuration. The baseline term BL is uncorrelated with every other variable, so its row and column are those of the identity. The biomarker $B$ couples only to $BR$, through $r$; its correlations with $TV$ and $PB$ are zero. The configurations differ in $A_c$, in $C_{cc'}$, and in $r$. The lower right $3n \times 3n$ submatrix is the response block $M$ of Section 4.2. The covariance matrix is $SRS$ with $S$ the diagonal matrix of standard deviations ($\sigma_{TV} = 10$, $\sigma_{PB} = 10\eta_t$ with $\eta_t$ the expectancy weight, $\sigma_{BR} = 8$ in the published parameters); because $S$ is positive diagonal, $SRS$ is positive definite exactly when $R$ is.

### A.2 Within-factor block $A_c$

**Compound symmetry (published; configurations A and D).** One constant for every pair of visits, irrespective of their separation:

$$[A_c]_{ts} = \begin{cases} 1, & t = s \\ \rho_{cs}, & t \neq s \end{cases} \qquad \Longleftrightarrow \qquad A_c = (1-\rho_{cs})I_n + \rho_{cs}J_n,$$

with $J_n$ the $n \times n$ matrix of ones. Its eigenvalues are $1 + (n-1)\rho_{cs}$ (once) and $1 - \rho_{cs}$ ($n-1$ times).

**AR(1) in calendar time (configurations B, C and E).** Correlation decays with the elapsed weeks between visits, not with the difference in visit index:

$$[A_c]_{ts} = \rho^{|w_t - w_s|}.$$

Unequally spaced designs therefore decay unequally: in the Hybrid design, whose visits fall at weeks 4, 8, 9, 10, 11, 12, 16 and 20, the one-week gaps of the discontinuation phase give adjacent correlations of 0.8, against $0.8^4 = 0.41$ across the four-week gaps.

### A.3 Cross-factor block $C_{cc'}$

Let $c_1$ be the same-visit cross-factor correlation (`c.cf1t`) and $c_\times$ the different-visit one (`c.cfct`).

**Constant (published; configurations A and D).**

$$[C_{cc'}]_{ts} = \begin{cases} c_1, & t = s \\ c_\times, & t \neq s \end{cases} \qquad \Longleftrightarrow \qquad C_{cc'} = c_\times J_n + (c_1 - c_\times)I_n.$$

**Decaying (configuration B; the current `covar`).** The same-visit entry is unchanged and the different-visit entries inherit the AR(1) decay:

$$[C_{cc'}]_{ts} = \begin{cases} c_1, & t = s \\ c_\times \rho^{|w_t - w_s|}, & t \neq s. \end{cases}$$

**Separable (configurations C and E).** The cross-factor block is the within-factor kernel scaled by $c_1$, so that

$$C_{cc'} = c_1 A, \qquad M = K \otimes A, \qquad K = (1-c_1)I_3 + c_1 J_3.$$

The three forms differ in what they imply for validity. The published response block decomposes into a person-level and an occasion-level term, both valid, and is positive definite for every schedule. The decaying form contains a same-visit term with no variance behind it and fails on dense schedules. The separable form is a single valid term and is positive definite for every schedule. The derivations are in `docs/35-separable-response-covariance.md`, Sections 4.4 and 4.5.

### A.4 Biomarker coupling vector $r$

All rules are written in terms of the residual fraction of the on-drug response at visit $t$,

$$\phi_t = \begin{cases} (1/2)^{t_{sd,t}/t_{1/2}}, & \text{visit } t \text{ off drug, after first exposure, } t_{1/2} > 0 \\ 0, & \text{otherwise,} \end{cases}$$

where $t_{sd,t}$ is the time since discontinuation.

**Step (published; configurations A and B).** The entry equals $c_{bm}$ wherever the carryover-adjusted $BR$ mean is nonzero and 0 elsewhere. Because that mean is nonzero off drug exactly when $\phi_t > 0$,

$$r_t = \begin{cases} c_{bm}, & \text{on drug} \\ c_{bm}\,\mathbf{1}[\phi_t > 0], & \text{off drug.} \end{cases}$$

**Graded (configuration C).** The off-drug entry is scaled by the residual fraction:

$$r_t = \begin{cases} c_{bm}, & \text{on drug} \\ c_{bm}\,\phi_t, & \text{off drug.} \end{cases}$$

**Mean moderation (configurations D and E).** $r = 0$: the biomarker writes no correlation into the matrix. Instead, after the draw, the $BR$ value at each on-drug visit is shifted by $\beta_{bm}\,\sigma_{BR}\,b$, where $b = (B - \mu_{bm})/\sigma_{bm}$ is the standardized biomarker and $\beta_{bm}$ is set equal to the nominal $c_{bm}$. The shift reproduces, on drug, the conditional mean that covariance moderation induces (paper 01, Section 2.2.1).

The step and graded rules switch the coupling on at the same visits. They differ only in its size off drug: a step to the full on-drug value as soon as any residual effect is present, against a value proportional to the residual.

### A.5 A worked example at the published parameters

Take four visits of Hybrid path A at weeks $w = (10, 11, 12, 16)$: on drug, off, off, on. The times since discontinuation are $t_{sd} = (0, 1, 2, 0)$, and the matrix of week gaps is

$$(|w_t - w_s|) = \begin{pmatrix} 0 & 1 & 2 & 6 \\ 1 & 0 & 1 & 5 \\ 2 & 1 & 0 & 4 \\ 6 & 5 & 4 & 0 \end{pmatrix}.$$

Set $\rho = 0.8$, $c_1 = 0.2$, $c_\times = 0.1$ and $c_{bm} = 0.3$. All entries below were computed (verified, `analysis/scripts/quick-sim/hendrickson-problems/06-worked-example-matrices.R`).

**Within-factor block $A_{br}$.**

$$A_{br}^{\mathrm{CS}} = \begin{pmatrix} 1.0000 & 0.8000 & 0.8000 & 0.8000 \\ 0.8000 & 1.0000 & 0.8000 & 0.8000 \\ 0.8000 & 0.8000 & 1.0000 & 0.8000 \\ 0.8000 & 0.8000 & 0.8000 & 1.0000 \end{pmatrix}, \qquad A_{br}^{\mathrm{AR(1)}} = \begin{pmatrix} 1.0000 & 0.8000 & 0.6400 & 0.2621 \\ 0.8000 & 1.0000 & 0.8000 & 0.3277 \\ 0.6400 & 0.8000 & 1.0000 & 0.4096 \\ 0.2621 & 0.3277 & 0.4096 & 1.0000 \end{pmatrix}.$$

Under compound symmetry the week-apart pair (1, 2) and the six-weeks-apart pair (1, 4) are equally correlated. Under AR(1) they are 0.80 and 0.26.

**Cross-factor block $C_{tv,pb}$.**

$$C^{\mathrm{constant}} = \begin{pmatrix} 0.2000 & 0.1000 & 0.1000 & 0.1000 \\ 0.1000 & 0.2000 & 0.1000 & 0.1000 \\ 0.1000 & 0.1000 & 0.2000 & 0.1000 \\ 0.1000 & 0.1000 & 0.1000 & 0.2000 \end{pmatrix}, \qquad C^{\mathrm{decaying}} = \begin{pmatrix} 0.2000 & 0.0800 & 0.0640 & 0.0262 \\ 0.0800 & 0.2000 & 0.0800 & 0.0328 \\ 0.0640 & 0.0800 & 0.2000 & 0.0410 \\ 0.0262 & 0.0328 & 0.0410 & 0.2000 \end{pmatrix},$$

$$C^{\mathrm{separable}} = \begin{pmatrix} 0.2000 & 0.1600 & 0.1280 & 0.0524 \\ 0.1600 & 0.2000 & 0.1600 & 0.0655 \\ 0.1280 & 0.1600 & 0.2000 & 0.0819 \\ 0.0524 & 0.0655 & 0.0819 & 0.2000 \end{pmatrix}.$$

The diagonal, the same-visit correlation $c_1$, is identical in all three. Off the diagonal, the constant form does not decay, the decaying form drops from 0.2 at the same visit to 0.08 one week later, and the separable form decays smoothly from 0.2 to 0.16. That drop in the decaying form is the same-visit excess that `docs/35` shows can make the response block indefinite. For this four-visit example the smallest eigenvalue of the response block is 0.100 (published), 0.029 (decaying) and 0.115 (separable).

**Coupling vector $r$.** The residual fractions at the two off-drug visits are $(\phi_2, \phi_3) = (2^{-1/t_{1/2}}, 2^{-2/t_{1/2}})$:

Table 15: Coupling vector $r$ under step, graded and mean-moderation rules by half-life

| $t_{1/2}$ | Step (A, B) | Graded (C) | Mean moderation (D, E) |
|---|---|---|---|
| 0 | (0.30, 0, 0, 0.30) | (0.30, 0, 0, 0.30) | (0, 0, 0, 0) |
| 0.1 | (0.30, 0.30, 0.30, 0.30) | (0.30, 0.0003, 0.0000, 0.30) | (0, 0, 0, 0) |
| 1.0 | (0.30, 0.30, 0.30, 0.30) | (0.30, 0.15, 0.075, 0.30) | (0, 0, 0, 0) |

At $t_{1/2} = 0$ the step and graded rules agree. At $t_{1/2} = 0.1$, where the residual drug effect is 0.1% one week after stopping and 0.0001% two weeks after, the step rule assigns the full 0.30 to both off-drug visits and the vector becomes constant, while the graded rule is indistinguishable from the no-carryover pattern. This is the discontinuity of Section 5.4, entry by entry. At $t_{1/2} = 1$ the graded entries halve and quarter, tracking the residual effect.

### A.6 The six configurations

Table 16: Components of the correlation matrix in each of the six configurations

| Component | A Published | B Adj. 1 | F Adj. 1 + 2, `covar` | C Adj. 1 + 2, separable | D Adj. 3 | E Adj. 1 + 3, separable |
|---|---|---|---|---|---|---|
| $A_c$ | CS | AR(1) | AR(1) | AR(1) | CS | AR(1) |
| $C_{cc'}$, $t \neq s$ | $c_\times$ | $c_\times \rho^{\text{gap}}$ | $c_\times \rho^{\text{gap}}$ | $c_1 \rho^{\text{gap}}$ | $c_\times$ | $c_1 \rho^{\text{gap}}$ |
| $r_t$ on drug | $c_{bm}$ | $c_{bm}$ | $c_{bm}$ | $c_{bm}$ | 0 | 0 |
| $r_t$ off drug | $c_{bm}\mathbf{1}[\phi_t > 0]$ | $c_{bm}\mathbf{1}[\phi_t > 0]$ | $c_{bm}\phi_t$ | $c_{bm}\phi_t$ | 0 | 0 |
| $BR$ mean shift on drug | none | none | none | none | $\beta_{bm}\sigma_{BR}b$ | $\beta_{bm}\sigma_{BR}b$ |
| Response block valid for every schedule | yes | no | no | yes | yes | yes |
| Ceiling on $c_{bm}$ | yes | yes | yes | yes | none | none |

Configurations F and C differ only in the cross-factor block, so their comparison isolates separability (Section 6.4). In the power simulation every configuration keeps the published values of everything else: the same-visit cross-factor correlation $c_1$, a common $\rho$ for the three factors (so the loop-order dependence noted in paper 01 Appendix B.5.2 does not arise), the Gompertz mean trajectories, the standard deviations, and the published recursive carryover on the $BR$ mean. Paper 01's own `covar` differs from configuration F in using $\rho = 0.7$ and the corrected, anchored carryover on the mean; neither affects the structure shown here.

### A.7 The separable structure, factorized

This section works the separable structure of configurations C and E through the four-visit example of A.5, to show concretely what the factorization does: the $12 \times 12$ response block splits into a $3 \times 3$

factor matrix and a $4 \times 4$ time matrix, every quantity needed (inverse, Cholesky factor, determinant, ceiling) is assembled from the two small pieces, and the time piece has closed-form factors that need no matrix computation at all. The parameters are those of A.5: $\rho = 0.8$, $c_1 = 0.2$, $c_{bm} = 0.3$, visits at weeks $(10, 11, 12, 16)$, on drug at the first and last. All entries were computed (verified, `analysis/scripts/quick-sim/hendrickson-problems/08-separable-factorization-example.R`).

### A.7.1 The factor part $K$

$$K = \begin{pmatrix} 1 & 0.2 & 0.2 \\ 0.2 & 1 & 0.2 \\ 0.2 & 0.2 & 1 \end{pmatrix}, \qquad K^{-1} = \begin{pmatrix} 1.0714 & -0.1786 & -0.1786 \\ -0.1786 & 1.0714 & -0.1786 \\ -0.1786 & -0.1786 & 1.0714 \end{pmatrix}, \qquad L_K = \begin{pmatrix} 1.0000 & 0 & 0 \\ 0.2000 & 0.9798 & 0 \\ 0.2000 & 0.1633 & 0.9661 \end{pmatrix}.$$

$K$ has eigenvalues 1.4 (once) and 0.8 (twice), so it is positive definite. Its inverse has the closed form $K^{-1} = (1-c_1)^{-1}(I_3 - \frac{c_1}{1+2c_1} J_3)$, and the entry that matters for the ceiling is $[K^{-1}]_{33} = (1+c_1)/((1-c_1)(1+2c_1)) = 1.0714 = 1/(1-R^2)$, where $R^2 = 0.067$ is the squared multiple correlation of $BR$ on $TV$ and $PB$ at the same visit. $L_K$ is the lower Cholesky factor, $K = L_K L_K^\top$.

**A.7.2 The time part $A$** The kernel $A$ is the AR(1) matrix of A.5. Its gaps are $d = (1, 1, 4)$ weeks, so the one-step correlations are $\rho_t = \rho^{d_t} = (0.8, 0.8, 0.4096)$ and the innovation scales are $\sqrt{1-\rho_t^2} = (0.6, 0.6, 0.9123)$. Its eigenvalues are 2.693, 0.825, 0.339 and 0.143.

Its inverse is tridiagonal: visits more than one step apart are conditionally independent given the visits between them.

$$A^{-1} = \begin{pmatrix} 2.7778 & -2.2222 & 0 & 0 \\ -2.2222 & 4.5556 & -2.2222 & 0 \\ 0 & -2.2222 & 2.9794 & -0.4922 \\ 0 & 0 & -0.4922 & 1.2016 \end{pmatrix}.$$

It factors as $A^{-1} = L^\top L$ with a lower bidiagonal $L$ whose entries come straight from $\rho_t$: $L_{11} = 1$, $L_{tt} = (1-\rho_t^2)^{-1/2}$, $L_{t,t-1} = -\rho_t(1-\rho_t^2)^{-1/2}$:

$$L = \begin{pmatrix} 1.0000 & 0 & 0 & 0 \\ -1.3333 & 1.6667 & 0 & 0 \\ 0 & -1.3333 & 1.6667 & 0 \\ 0 & 0 & -0.4490 & 1.0962 \end{pmatrix}, \qquad L_A = \begin{pmatrix} 1.0000 & 0 & 0 & 0 \\ 0.8000 & 0.6000 & 0 & 0 \\ 0.6400 & 0.4800 & 0.6000 & 0 \\ 0.2621 & 0.1966 & 0.2458 & 0.9123 \end{pmatrix}.$$

$L_A$ is the Cholesky factor of $A$ itself ($A = L_A L_A^\top$, and $L_A = L^{-1}$). Row $t$ of $L$ is the AR(1) recursion read backwards: $(LX)_t = (X_t - \rho_t X_{t-1})/\sqrt{1-\rho_t^2}$ is the standardized innovation at visit $t$. The agreement $L^\top L = A^{-1}$ holds to $9 \times 10^{-16}$.

**A.7.3 Assembling the response block** With factor-major ordering the $12 \times 12$ response block is

$$M = K \otimes A = \begin{pmatrix} A & 0.2A & 0.2A \\ 0.2A & A & 0.2A \\ 0.2A & 0.2A & A \end{pmatrix},$$

each off-diagonal block being the separable $C^{\text{separable}}$ of A.5. Every quantity of $M$ is then built from the two small pieces (verified to rounding error):

Table 17: Full and factored computations of the separable response block compared

| Quantity | Full $12 \times 12$ computation | Factored form | Agreement |
|---|---|---|---|
| Inverse | $M^{-1}$ | $K^{-1} \otimes A^{-1}$ | $9 \times 10^{-16}$ |
| Cholesky factor | $\mathrm{chol}(M)$ | $\mathrm{chol}(K) \otimes \mathrm{chol}(A)$ | $2 \times 10^{-16}$ |
| Determinant | $\det M = 8.0867 \times 10^{-4}$ | $(\det K)^4 (\det A)^3$ | exact |
| Eigenvalues | 12 values | products $\omega_i(K)\,\omega_j(A)$ | by construction |

The last row is the validity guarantee: every eigenvalue of $M$ is a product of a positive eigenvalue of $K$ and a positive eigenvalue of $A$, whatever the visit times.

**A.7.4 The ceiling, factored** With the biomarker coupled to $BR$ through the pattern $g$,

$$c_{bm}^{*} = \left([K^{-1}]_{33}\, g^{\top} A^{-1} g\right)^{-1/2}, \qquad g^{\top} A^{-1} g = g_1^2 + \sum_{t \geq 2} \frac{(g_t - \rho_t g_{t-1})^2}{1 - \rho_t^2}.$$

The sum runs over the three steps between consecutive visits, so the ceiling can be read as a cost per step. For three coupling patterns on this example:

Table 18: Per-step decomposition of the ceiling for three coupling patterns

| Coupling $g$ | $g_1^2$ | Step 1 | Step 2 | Step 3 | $g^{\top} A^{-1} g$ | Ceiling (closed form) | Ceiling (full matrix) |
|---|---|---|---|---|---|---|---|
| No carryover $(1, 0, 0, 1)$ | 1.000 | 1.778 | 0.000 | 1.202 | 3.979 | 0.4843 | 0.4843 |
| Graded, $t_{1/2} = 1$: $(1, 0.5, 0.25, 1)$ | 1.000 | 0.250 | 0.063 | 0.968 | 2.281 | 0.6397 | 0.6397 |
| Step, any carryover $(1, 1, 1, 1)$ | 1.000 | 0.111 | 0.111 | 0.419 | 1.641 | 0.7541 | 0.7541 |

These are ceilings for this four-visit fragment, not for the full eight-visit design. The table shows where the cost lies. Without carryover the on-to-off step across a one-week gap costs $\rho_t^2/(1-\rho_t^2) = 1.778$, and the off-to-on step across the four-week gap costs $1/(1-\rho_t^2) = 1.202$; those two switches set the ceiling. Graded coupling softens both switches. The step rule's constant vector pays almost nothing, because it has no switches, which is why its ceiling is the highest of the three and also why it carries no on-drug versus off-drug contrast.

**A.7.5 The factored draw** Arranging one participant's response as a $4 \times 3$ matrix $\mathbf{X}$, one column per factor, the separable structure is the matrix-normal model $\mathbf{X} = L_A \mathbf{Z} L_K^{\top}$ with $\mathbf{Z}$ standard normal. Neither factor needs to be computed as a matrix:

1. **Time.** Each column of $L_A\mathbf{Z}$ is an AR(1) path, generated by $H_1 = z_1$ and $H_t = \rho_t H_{t-1} + \sqrt{1-\rho_t^2}\, z_t$, here $H_2 = 0.8H_1 + 0.6z_2$, $H_3 = 0.8H_2 + 0.6z_3$ and $H_4 = 0.4096H_3 + 0.9123z_4$.
2. **Factors.** Multiplying by $L_K^{\top}$ blends the three paths: $TV = H^{(1)}$, $PB = 0.2H^{(1)} + 0.9798H^{(2)}$, $BR = 0.2H^{(1)} + 0.1633H^{(2)} + 0.9661H^{(3)}$.
3. **Biomarker.** Given the response, $B$ is normal with mean $\sum_{c,t} W_{ct} X_{c,t}$ and variance $1 - c_{bm}^2 q$, where $W = c_{bm}\,[K^{-1}]_{\cdot 3}\,(A^{-1}g)^{\top}$ and $q = [K^{-1}]_{33}\, g^{\top} A^{-1} g$. For the graded pattern at $t_{1/2} = 1$:

$$W = \begin{pmatrix} -0.0893 & 0.0268 & 0.0460 & -0.0578 \\ -0.0893 & 0.0268 & 0.0460 & -0.0578 \\ 0.5357 & -0.1607 & -0.2759 & 0.3467 \end{pmatrix} \begin{matrix} \leftarrow TV \\ \leftarrow PB, \\ \leftarrow BR \end{matrix} \qquad 1 - c_{bm}^2 q = 0.7801.$$

The weights on $BR$ are largest at the on-drug visits, as expected. The small negative weights on $TV$ and $PB$ remove the part of $BR$ that is shared with them, since the biomarker is correlated with $BR$ only. The residual variance 0.7801 is positive because $c_{bm} = 0.3$ is below this fragment's ceiling of 0.6397. At $c_{bm} = 0.6397$ it would be exactly zero, and above it there is no distribution to draw from.

**A.7.6 Why the other two structures do not factor this way** **The current `covar` block** decomposes into two terms, one of them indefinite (verified):

$$M_{\text{covar}} = K_a \otimes A + K_b \otimes I_4, \qquad K_a = \begin{pmatrix} 1 & 0.1 & 0.1 \\ 0.1 & 1 & 0.1 \\ 0.1 & 0.1 & 1 \end{pmatrix}, \qquad K_b = \begin{pmatrix} 0 & 0.1 & 0.1 \\ 0.1 & 0 & 0.1 \\ 0.1 & 0.1 & 0 \end{pmatrix}.$$

$K_b$ has eigenvalues 0.2, $-0.1$ and $-0.1$. Because the two terms carry different time kernels ($A$ and $I_4$), $M_{\text{covar}}$ is not a single Kronecker product and its inverse does not split. Its validity depends on the schedule: here $\omega_{\min}(A) = 0.1433$ against the threshold $(c_1 - c_\times)/(1 - c_\times) = 0.1111$, so this fragment is valid, and the predicted smallest eigenvalue of $M_{\text{covar}}$, $\omega_{\min}(A)(1 - c_\times) - (c_1 - c_\times) = 0.0290$, matches the computed one. A schedule with closer visits would push $\omega_{\min}(A)$ below the threshold.

**The published block** decomposes into two valid terms (verified):

$$M_{\text{orig}} = K_o \otimes I_4 + K_p \otimes J_4, \qquad K_o = \begin{pmatrix} 0.2 & 0.1 & 0.1 \\ 0.1 & 0.2 & 0.1 \\ 0.1 & 0.1 & 0.2 \end{pmatrix}, \qquad K_p = \begin{pmatrix} 0.8 & 0.1 & 0.1 \\ 0.1 & 0.8 & 0.1 \\ 0.1 & 0.1 & 0.8 \end{pmatrix},$$

an occasion-level term and a person-level term with eigenvalues (0.4, 0.1, 0.1) and (1.0, 0.7, 0.7). Both $K_o$ and $K_p$ are positive definite, so $K_o \otimes I_4$ is positive definite and $K_p \otimes J_4$ positive semidefinite ($J_4$ has rank one), and the published response block is valid for every schedule (smallest eigenvalue 0.100 here), but it too is a sum of two Kronecker products rather than one, and it has no decay with lag.